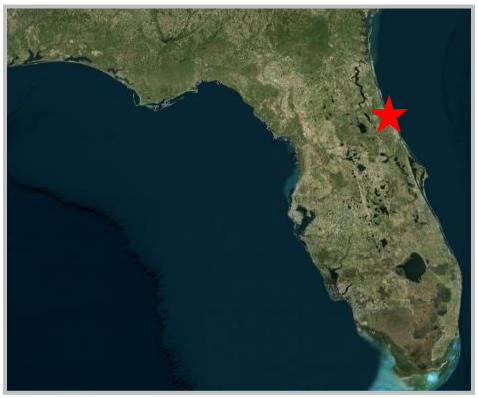
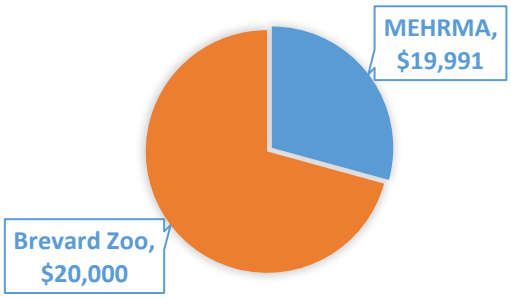
# Oyster Reef Restoration, Volusia County, FL






**Partners**

- Brevard Zoo, University of Central Florida

**Grant support**

Brevard Zoo has an IRLNEP grant for $65,675 (with $65,675 provided in match) supporting their oyster recycling and reef restoration program. The MEHRMA funds and associated match allowed the program to be expanded to additional reefs in the Indian River Lagoon system. The Zoo staff conducted numerous volunteer events to construct and deploy the mats used for this project, thereby providing volunteer time as match.

**Project Description**

Brevard Zoo has teamed with UCF to restore oyster reefs in Mosquito Lagoon over the past few years. The reefs suffer from boat wakes piling live oysters above the intertidal zone, killing the oysters and creating "dead margins" of shell on once productive reefs. Volunteers assist by raking down the dead margins to intertidal elevations, and deploying weighted oyster mats on the affected reef areas. Once in place, oyster larvae quickly recruit to the shells and grow to make a functional reef system. The mats are created using oysters attached to mats by volunteers, including school groups and civic organizations. In this manner, over 75 reefs have been restored and secured in Mosquito Lagoon, Volusia County. UCF monitors the restored reef, tracking performance measures such as oyster recruitment, growth, and survivorship.

**Objectives**

The purpose of this project was to create and deploy 1000 oyster mats for three dead margin reefs in the Mosquito Lagoon, thereby securing the affected areas. The shells serve as substrate for the recruitment and colonization of oyster larvae. The enhanced habitat provides nursery and foraging habitat for a variety of wildlife including fish, crabs, shrimps, other mollusks, and wading birds.

**Benefits**

- Attenuate waves
- Reduce the effects of boat wakes
- Provide substrate for oyster larval settlement

- Provide nursery and foraging habitat for a variety of wildlife including fish, crabs, shrimps, other mollusks, and wading birds

**FWC Strategic Plan Elements**

This project directly addresses the following FWC Strategic Plan Elements:

*Theme I, Goal 2: Ensure sufficient habitats exist to support healthy and diverse fish and wildlife populations.*

*Strategies:*

2. *Protect lands and waters critical to sustaining healthy and diverse fish and wildlife populations through diverse incentive programs.*
3. *Manage habitats to sustain healthy and diverse fish and wildlife populations.*

**Acreage restored by habitat type (footprint)**

- 1000 oyster mats (0.06 acres) placed to:
  - restore and secure dead margins along three affected oyster reefs
  - enhance 0.25 acres of oyster habitat by reducing shell movement and by providing indirect benefits through improved water quality and habitat

**Acreage affected by restoration activities**

- Total project acreage is 0.25 acres

**Project Map (identify acreage of both footprint and affected area)**

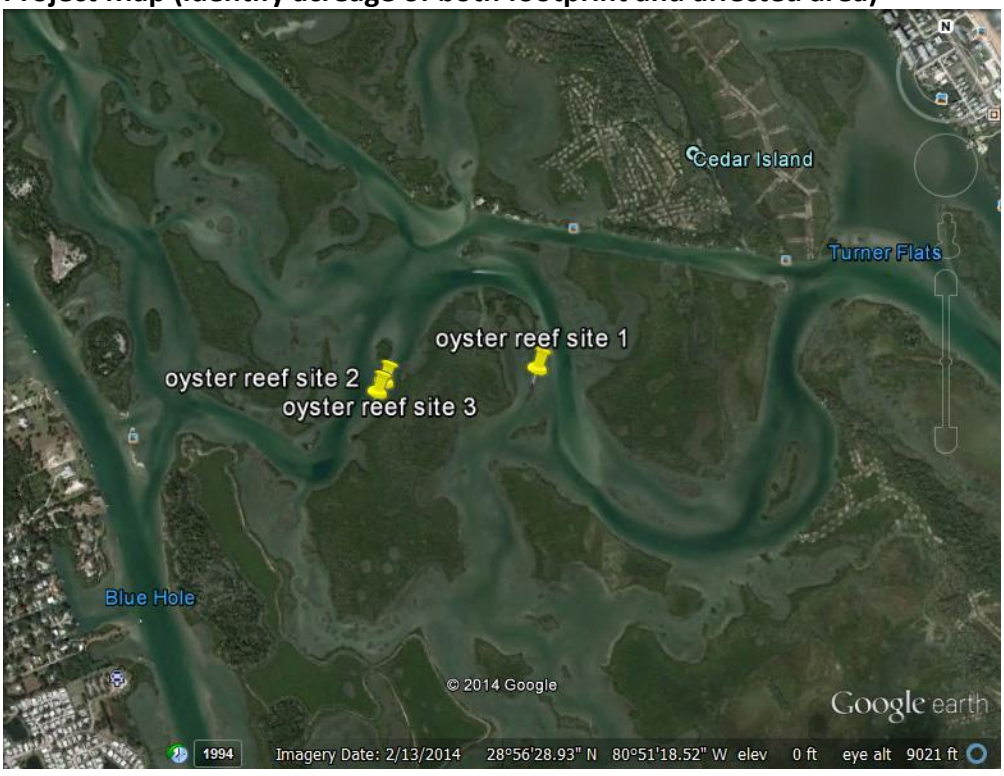


Photo 1: Three dead margin sites in Mosquito Lagoon restored via MEHRMA funding 2014-15



Photo 2: Volunteers assemble oyster mats



Photo 3: Mats ready for deployment



Photo 4: Volunteers deploy mats; restored reef (L) versus natural reef (R)




Photo 5: Restored reef in Mosquito Lagoon; oyster mats with live oysters